AI & SOCIETY COURSE

Prof. Mor Naaman

Part 5

AI'S WORLDWIDE IMPACT

Unit 1: AI and Work

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Where We Are in the Course

Part 1 — AI Foundations
1. What is Generative AI: How it Works and What it Can Do
2. Bias and Limitations: Where GenAI Fails

Part 2 — AI and Individuals
3. AI and You: Cognition, Reliance, and Creativity
4. Persuasion by Design: How AI Influences You

Part 3 — AI, Communication, and Information
5. AI-Mediated Communication and Relationships
6. AI, Culture, and the Information Environment

Part 4 — Challenges for AI Implementation
7. Trust and Safety: AI Harms and Prevention
8. Data, Privacy, and IP: Who Owns What
9. The Challenging Economics of AI

Part 5 — AI's World-wide Impact
10. AI and the World: Education, Work, Power, Democracy
11. Externalities: Human and Environmental Costs of AI

Part 6 — The Way Forward with AI
12. The Future: Navigating AI in Your Community

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This unit: goals

- Understand how generative AI is actually being used at work today
- See what the evidence says about AI and productivity
- Understand the two types of impact on work: does AI makes work easier, or AI takes the work?

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How is generative AI being **currently used** in the workplace?

(still changing all the time!)

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Where is biggest use of AI for business?

- Fastest growing: coding.
  - Code generation is the fastest-growing enterprise GenAI use case
  - Huge improvements in 2026; coding agents like Claude Code capable of multi-file repository changes, advanced architectural reasoning, end-to-end development tasks.
- Other key categories:
  - Sales/marketing
  - Customer service
  - Finance "back office" (example: processing invoices)
- Still open question: what is the value at scale, over time, for different types of companies.

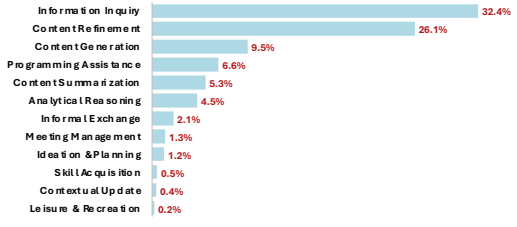
Sources: Menlo Ventures, 2025: The State of Generative AI in the Enterprise; McKinsey, State of AI 2026.

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What are people using it for at work (Microsoft)

Microsoft, Share of M365 Copilot messages by topic — Feb 2026



Source: Counts et al. (2026), "AI in the Enterprise"

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Does using generative AI increase **productivity?**

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One early study

Experimental Evidence on the Productivity Effects of Generative Artificial Intelligence

Shakked Noy MIT
Whitney Zhang MIT

March 2, 2023

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Method

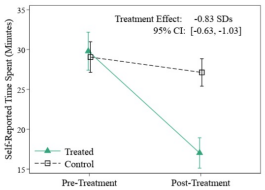
- Recruited 444 subjects (marketers, grant writers, consultants, data analysts, human resource professionals, managers) in 2023
- Asked them to do occupation-specific task, eg writing press releases, short reports, analysis plans, and delicate emails
  - 20-to 30-minute assignments designed to resemble real tasks performed in these occupations
- Gave half of them access to ChatGPT for a second round task.

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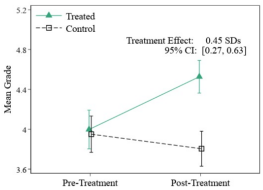
Less time, higher scores

(a) Time Taken Decreases



40% decrease in time taken

(b) Average Grades Increase

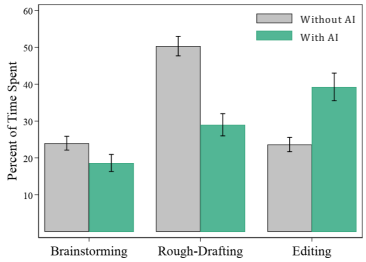


18% increase in work quality

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Save a lot of time drafting (but a bit more editing)



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Shifting Work Patterns with Generative AI*

Eleanor Wiske Dillon† Microsoft Research
Sonia Jaffe† Microsoft Research
Nicole Immorlica Microsoft Research
Christopher T. Stanton Harvard Business School, NBER, and CEPR

May 6, 2025

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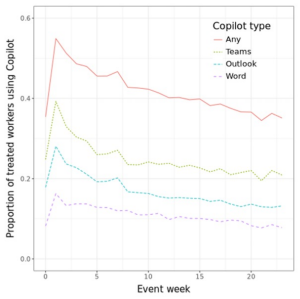
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Methods

- Co-Pilot: Microsoft's AI (their installation of ChatGPT)
- 7137 workers spanning many occupations that involve heavy use of Microsoft Office tools into which Copilot was integrated (e.g., emailing, video meetings, chat, document creation)
- During early rollout of Copilot (September 2023)
- Ran for 6 months
- Observe their use of Microsoft Office products

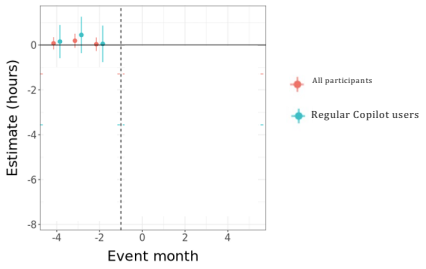
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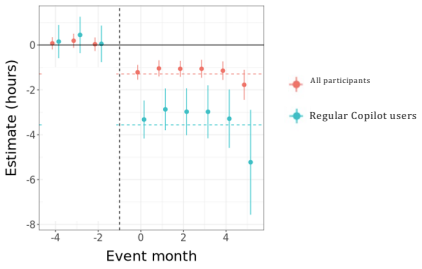
(a) Time in Outlook

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On average, workers with access to Copilot spent 1.3 hours less per week working on email.

Workers who used Copilot *regularly* spent 3.6 fewer hours per week on email.



(a) Time in Outlook

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What about productivity in writing code?

Potential benefits:

Faster task completion

Higher code output volume

Build new products and features faster

But, potential drawbacks:

Potentially more bugs, less efficient code

Engineers need to review huge amounts of AI code

Who will do output verifications and quality control?

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What about productivity in writing code?

- This field is changing fast! Not a lot of recent research.
- One recent paper (study done 2025):

"While frequent GenAI users reported faster task completion and higher output volume, these gains were offset by increased code review burden, persistent cognitive load from output verification, …"

- But: that was before the 2026 jump in coding abilities!

Source: Afroz et al. (2026), "The Fast and Spurious: Developer Productivity with GenAI"

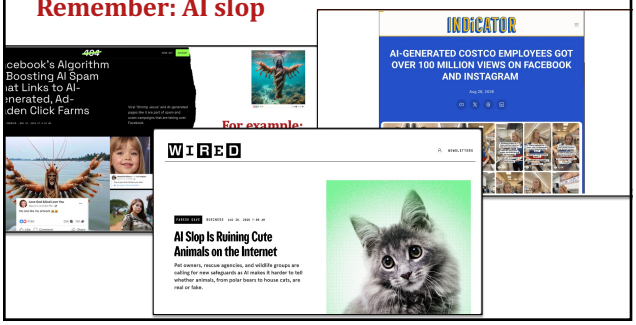
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Productivity or slop?

- Many tasks: AI seems to be increasing output, but it not actually ..

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Remember: AI slop



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Danger: Workslop

Workslop *(n.)*
Low-quality, AI-generated content that looks professional at first glance but carries little substance especially considering its length.

Source: Niederhoffer et al. (2025). AI-Generated "Workslop" Is Destroying Productivity

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Danger: Workslop

Workslop *(n.)*
Low-quality, AI-generated content that looks professional at first glance but carries little substance especially considering its length.

Instead of creating valuable work (e.g. code, documents, presentations) we get people creating AI content, transferring the load and work onto the colleague who receives it and needs to evaluate it.

In one paper, 41% of surveyed workers report having received Workslop.

Source: Niederhoffer et al. (2025). AI-Generated "Workslop" Is Destroying Productivity

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Summary so far

- AI can increase productivity in many tasks – coding, summarization, documentation
- Danger of Workslop: too easy to create a lot of AI content that is hard to review, verify, check.
- My recommendations:
  - Human boundaries on production: make sure somebody has the time and capacity to verify, evaluate, review
  - Responsibility: when you create something with AI, you verified, understand, and know everything that is in it
  - Accountability: when you create something with AI you are accountable for outcomes resulting from it or its content

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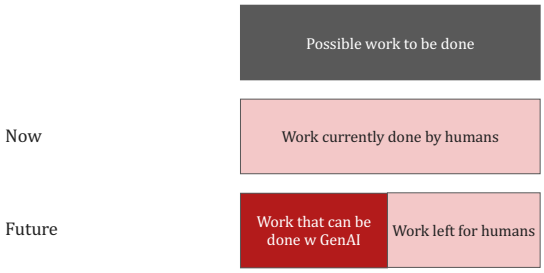
Two narratives

AI is going to make your job **easier** / make you **more productive**

AI is going to **take** your job

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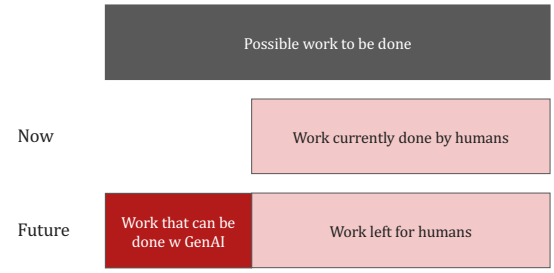
Scenario A



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Scenario B



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Better question: which jobs are more like Scenario A vs B?

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Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

Erik Brynjolfsson* Bharat Chandar† Ruyu Chen‡§¶

August 26, 2025

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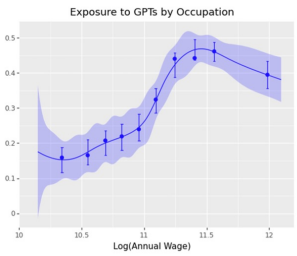
Methods

- Uses data from ADP, a firm in America that helps companies process payments to their workers (over 25 million workers in the US)
- Tracks employment changes for workers in occupations measured as more or less exposed to artificial intelligence

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Eloundou et al 2024



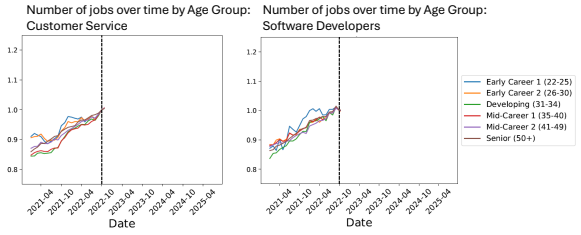
Occupations with highest exposure	% Exposure
Interpreters and Translators	76.5
Survey Researchers	75.0
Poets, Lyricists and Creative Writers	68.8
Animal Scientists	66.7
Public Relations Specialists	66.7
Survey Researchers	84.4
Writers and Authors	82.5
Interpreters and Translators	82.4
Public Relations Specialists	80.6
Animal Scientists	77.8
Mathematicians	100.0
Tax Preparers	100.0
Financial Quantitative Analysts	100.0
Writers and Authors	100.0
Web and Digital Interface Designers	100.0

(note: exposure estimates are based on the opinions of the people writing the paper, not actual data)

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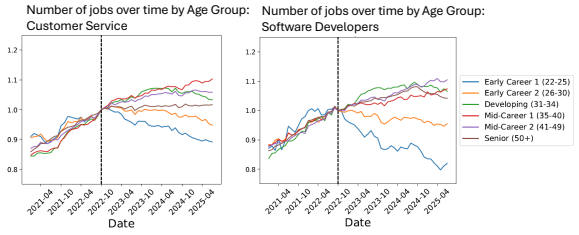
Two particularly exposed occupations



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Two particularly exposed occupations

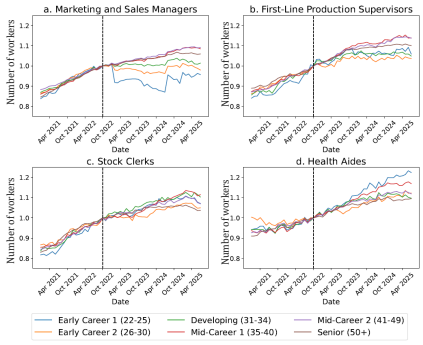


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Highest exposure

Lowest exposure



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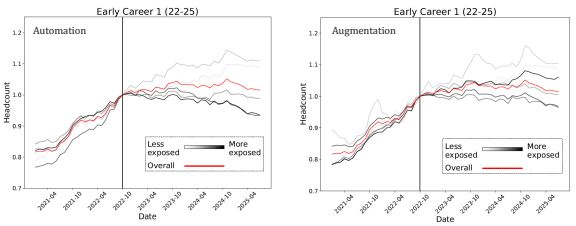
Automation vs augmentation

- Anthropic Economic Index estimates fraction of Claude queries related to each occupation
- Also estimates if each task is automation (AI replacing human) or augmentation (AI helping human) or neither
- So we can break out changes in early career employment by whether a given occupation has mostly automation vs augmentation queries

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Automation vs augmentation



Conclusion: Early career hiring is down in high-AI automation jobs but not high-AI augmentation jobs

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Summary

- Declines in employment for early-career workers in occupations most exposed to AI, such as software development and customer support
- Entry-level employment has declined in applications of AI that *automate* work, not so much effects for those that *augment* it
- Everything still changing!

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This unit: summary

- Individual use of AI at work is widespread; enterprise adoption growing
- Likely productivity improvements for well-defined writing and support tasks
- But: be careful of Workslop!
- Coding productivity impact: likely significant now, but who is evaluating all this code?
- Overall employment effects will differ: automation vs. augmentation

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